FARM MANAGEMENT SYSTEM

Salesforce CRM Final Project Report

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A Salesforce-Based Agricultural Data Management Solution

Project Documentation

Bachelor of Computer Applications (BCA)

Salesforce Developer Edition / Lightning Experience

1. PROJECT OVERVIEW

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. The system centralizes information about farms, crops, farmers, and buyers in one cloud platform. It supports structured data entry, validation, relationships between records, role-based access, reports, and dashboards.

1.1 Project Objectives

- Digitize farm and agricultural records and reduce dependence on manual paperwork.
- Maintain centralized and accessible information about farms, crops, farmers, and buyers.
- Improve data accuracy through structured fields, relationships, and validation.
- Provide real-time reports and dashboard visualizations for monitoring.
- Support role-based access and secure storage of business data.
- Improve reporting speed and help users make data-driven decisions.

1.2 User Story

As a farmer, farm manager, or system administrator, I want a centralized Salesforce platform to record and manage farm, crop, farmer, and buyer information so that records can be updated easily, validated consistently, analyzed through reports and dashboards, and accessed according to user roles.

2. PURPOSE AND SCOPE

2.1 Purpose

The purpose of the project is to digitize agricultural data, improve accuracy and accessibility, automate record handling, enable real-time monitoring, and support data-driven decisions. By storing information in Salesforce Cloud, the system provides a centralized and scalable environment for agricultural management.

2.2 Scope

- Farm record management
- Crop record management
- Farmer registration and farm relationship management
- Buyer information management
- Custom tabs and Lightning application navigation
- Field relationships and field dependencies
- Profiles, roles, users, and sharing rules
- Reports and dashboard visualization
- Testing and validation of major system functions

3. PROBLEM STATEMENT AND IDEATION

3.1 Problem Statement

User	Problem	Expected Improvement
Farmer	Manual or incomplete farm and crop records.	Centralized and accurate farm information.

Farm Manager	Delayed or inaccurate reports.	Real-time reports and dashboard monitoring.
Buyer	Limited access to updated crop availability and related information.	Better access to current records.
System Administrator	Manual access control and record administration.	Role-based profiles, roles, users, and sharing rules.

3.2 Empathy Map

The target users are Farmers, Farm Managers, Buyers, and System Administrators. Their common needs include accurate records, easy access, faster reporting, secure data, and better visibility into farm operations.

3.3 Brainstormed Ideas

- Manual paper-based records
- Excel-based agricultural records
- A general web-based farm management system
- A mobile farm management application
- A Salesforce-based Farm Management System

The Salesforce-based approach was selected as the proposed solution because the project requires centralized cloud data, structured relationships, access control, reporting, and dashboards.

4. REQUIREMENTS AND SOLUTION ARCHITECTURE

4.1 Customer/User Journey

1. User logs in to Salesforce.
2. User accesses the Farm Management System application.
3. Farm and crop data are entered or updated.
4. Farmers and buyers are registered.
5. Reports are generated from stored records.
6. Users analyze the information through reports and dashboards.
7. The information supports operational decision-making.

4.2 Functional Requirements

ID	Requirement	Description
FR-1	Authentication	Users should securely access the Salesforce application.
FR-2	Data Management	Users should manage Farm, Crop, Farmer, and Buyer records.
FR-3	Data Validation	Entered information should follow configured field and validation rules.
FR-4	Relationship Mapping	Records should be connected using Salesforce relationships and dependencies.
FR-5	Reports	The system should generate reports from project data.

FR-6	Dashboards	The system should visualize selected information using dashboard components.
FR-7	User Access	Profiles, roles, users, and sharing rules should control access.

4.3 Non-Functional Requirements

- Usability
- Security
- Reliability
- Performance
- Availability
- Scalability

4.4 Data Flow / DFD

The high-level data flow is: User → Salesforce Lightning UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards.

4.5 Solution Architecture

Layer	Implementation
Presentation Layer	Salesforce Lightning UI
Application Layer	Validation Rules and Relationships
Data Layer	Custom Salesforce Objects
Analytics Layer	Salesforce Reports and Dashboards

5. TECHNOLOGY STACK AND IMPLEMENTATION PLAN

Technology Area	Technology Used
Frontend	Salesforce Lightning UI
Backend / Platform	Salesforce Platform
Database	Salesforce Cloud
Automation / Validation	Salesforce Validation Rules
Analytics	Salesforce Reports and Dashboards

5.1 Core Modules

- Farm Management
- Crop Management
- Farmer Management
- Buyer Management
- User and Security Management
- Reports and Dashboards

6. DEVELOPMENT AND SALESFORCE CONFIGURATION

6.1 Salesforce Developer Environment

The project is implemented in a Salesforce Developer Edition environment using Lightning Experience. Salesforce provides the cloud platform, object configuration, security controls, reporting, and dashboard capabilities required for the Farm Management System.

6.2 Custom Objects

Four custom objects form the core data model: Farm, Crop, Farmer, and Buyer. The project documentation uses these objects to store agricultural and business information.

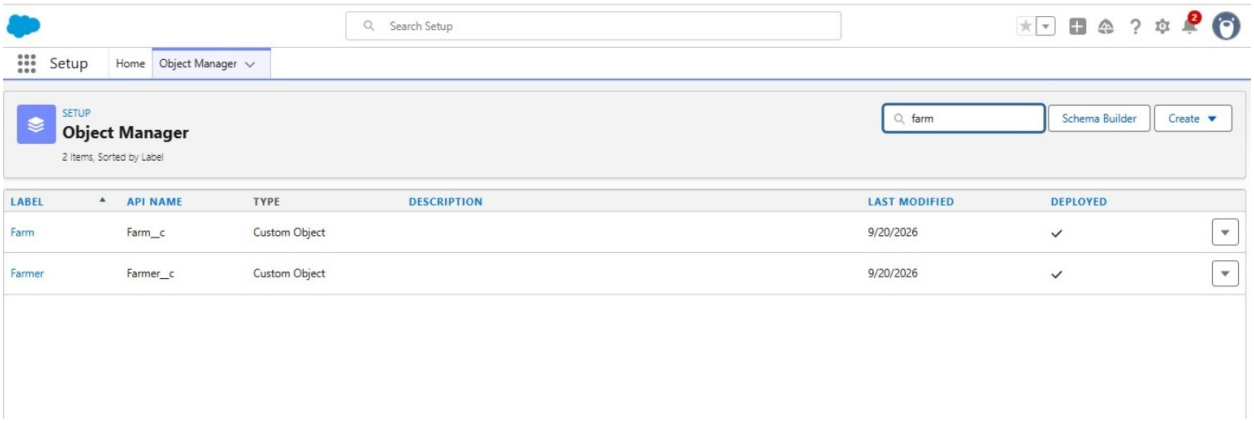


Figure 9. Salesforce Object Manager showing Farm and Farmer custom objects.

6.3 Farm Object

The Farm object stores farm-specific information. The configured fields include Farm Name, Farm Type, Farmer, Irrigation Type, Location, Size, and Soil Quality. The Farmer field is configured as a Master-Detail relationship.

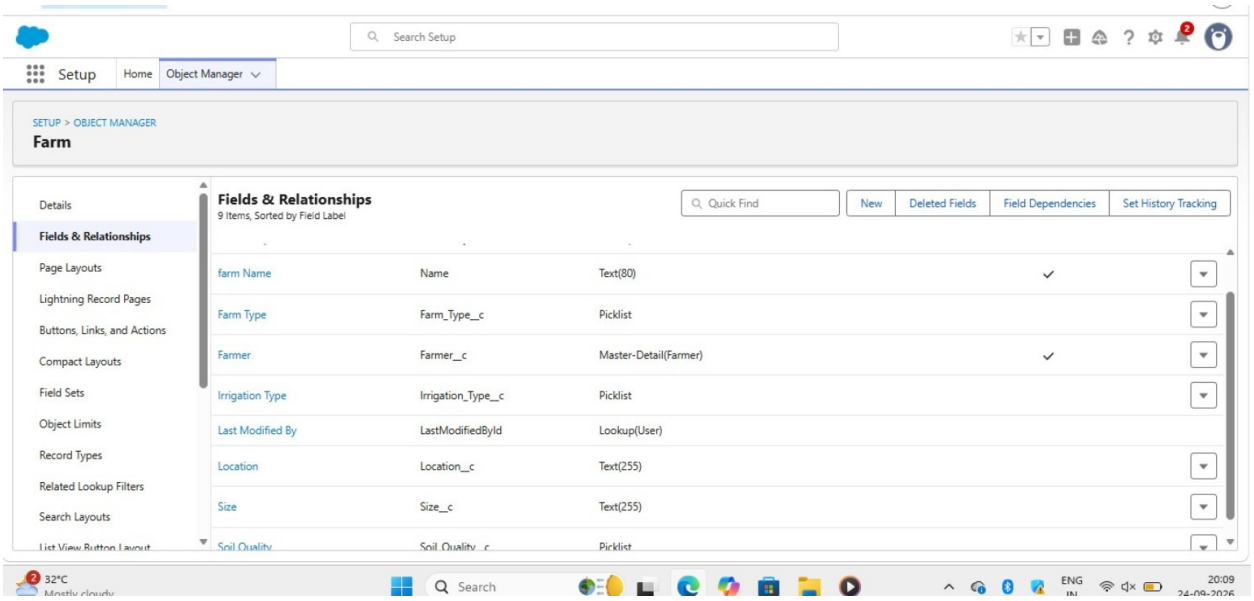


Figure 5. Farm object Fields & Relationships configuration.

6.4 Farm Fields and Relationships

- Farm Type: Crop, Livestock, Mixed.
- Soil Quality: Organic, Cover Crops, Composting.
- Irrigation Type: Sprinkler, Drip, Localized.
- Location: text field for farm location.
- Size: field for farm size.
- Farmer: Master-Detail relationship to Farmer.

6.5 Crop Object

The Crop object manages crop-related information including Crop Name, Planting Date, Harvest Date, Farm relationship, Actual Yield, and Total Acres Harvested. The project also defines a formula field for Expected Yield per Acre using Actual Yield divided by Total Acres Harvested.

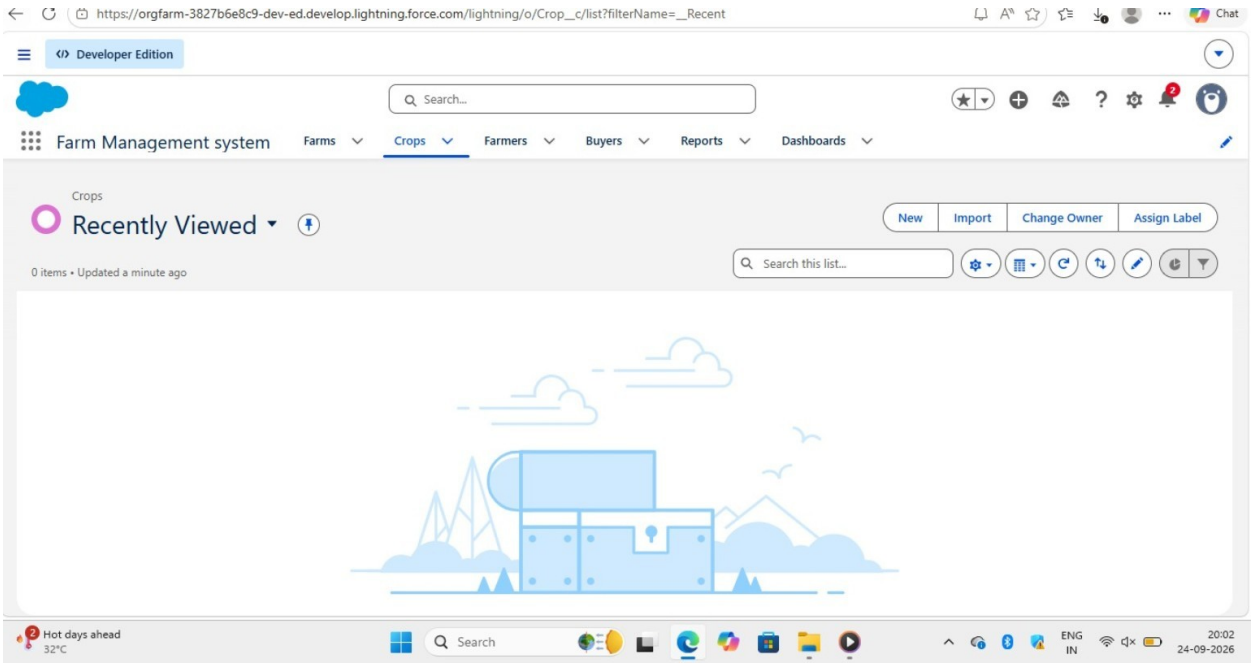


Figure 10. Farm Management System Crops tab.

6.6 Farmer Object

The Farmer object stores Contact Number, State, District, and Village. The project configures State values for Telangana, Andhra Pradesh, and Tamilnadu, with a field dependency between State and District.

6.7 Buyer Object

The Buyer object stores Contact, Email, Address, and Buyer Type. Buyer Type contains Wholesale and Retail options.

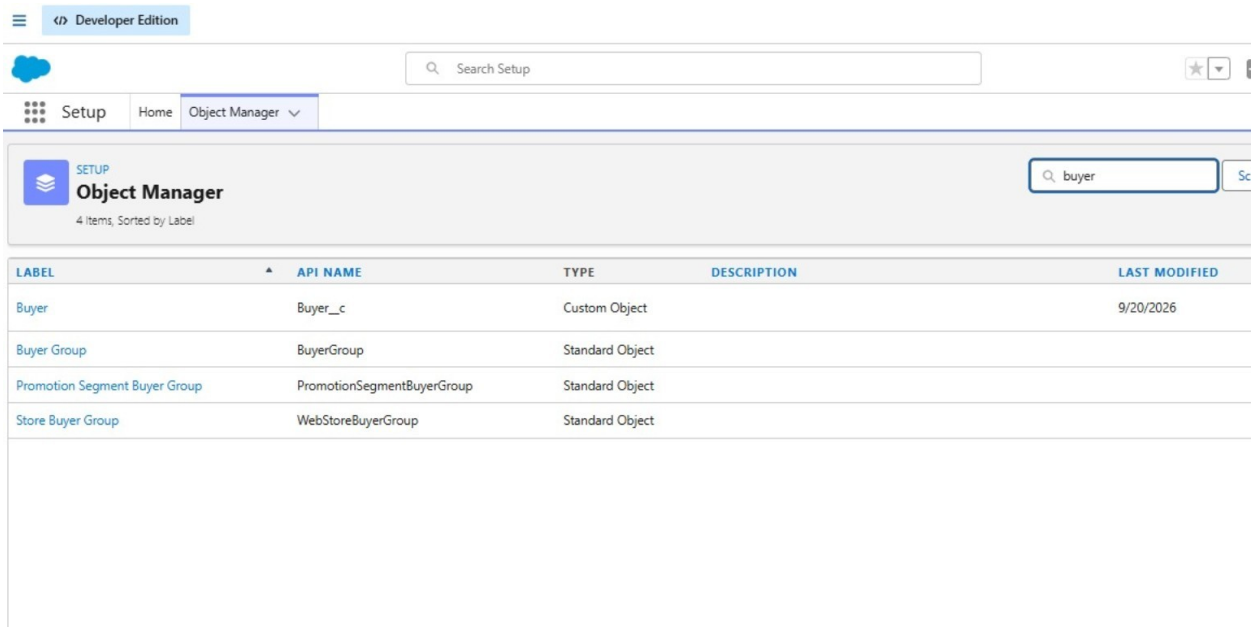


Figure 2. Salesforce Object Manager showing the Buyer custom object.

7. CUSTOM TABS AND LIGHTNING APPLICATION

7.1 Custom Tabs

Custom tabs were configured for the four project objects so users can access Farms, Crops, Farmers, and Buyers directly from the application.

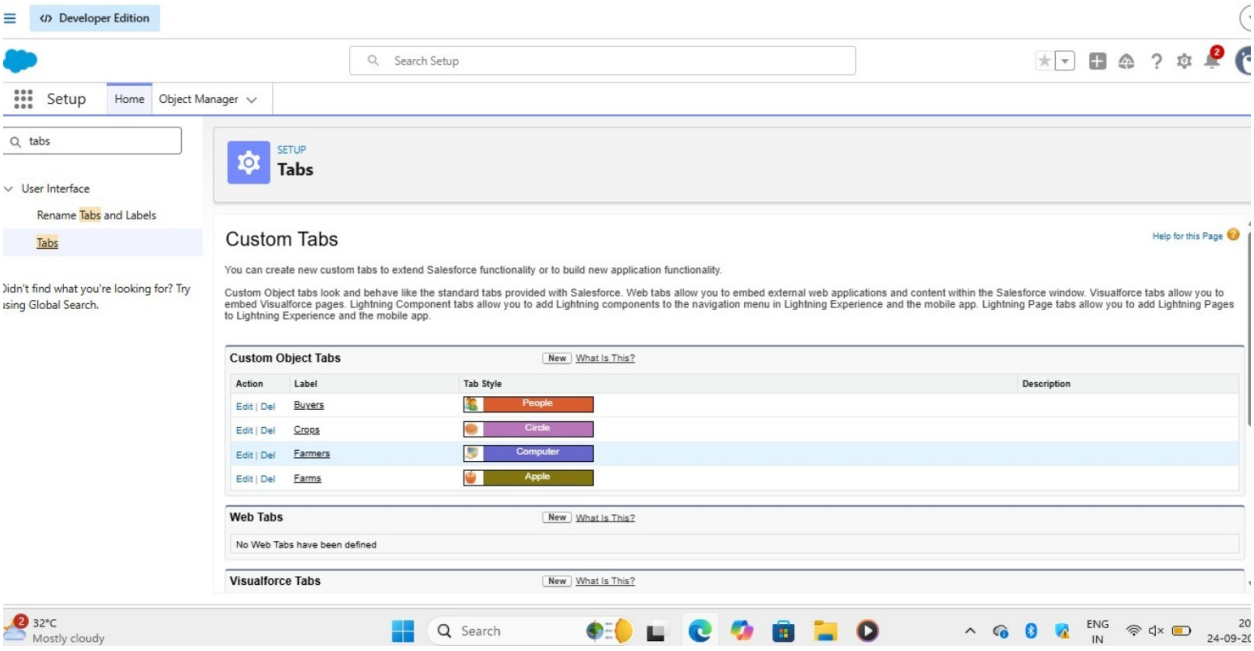


Figure 4. Custom Tabs configuration for Buyers, Crops, Farmers, and Farms.

7.2 Farm Management System Lightning App

The Farm Management System Lightning application provides navigation for Farms, Crops, Farmers, Buyers, Reports, and Dashboards. The application is intended to provide a centralized user interface for the project.

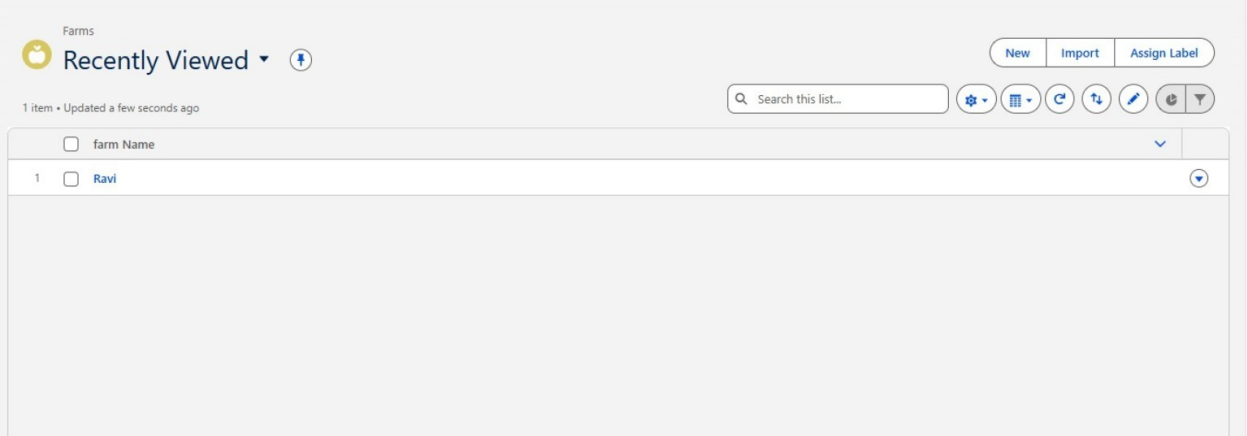


Figure 7. Farm Management System Farms tab with a recently viewed farm record.

8. SECURITY, PROFILES, ROLES AND SHARING

8.1 Profiles

The project creates custom profiles by cloning the Standard User profile. The documented profiles include Sales Department and Farm Manager.

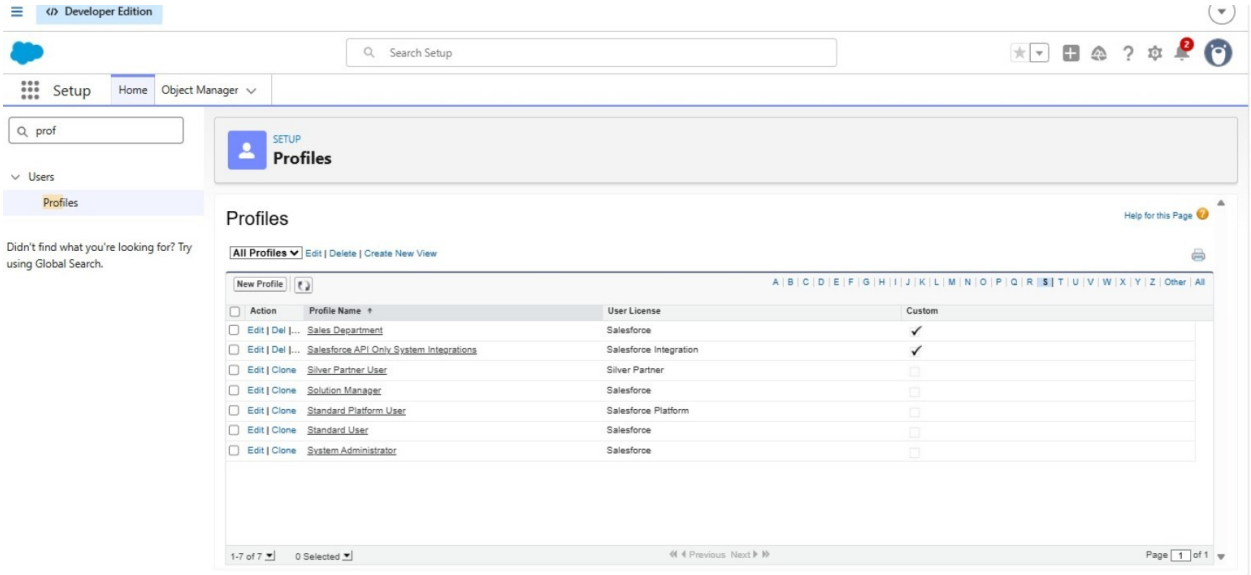


Figure 1. Salesforce Profiles configuration showing the Sales Department profile.

8.2 Role Configuration

The Farm Manager role is configured under the CEO role in the role hierarchy.

8.3 Users

The project creates a Farm Manager user using the Farm Manager role and Farm Manager profile with a Standard Platform User license. A Sales Department user is also configured.

8.4 Sharing Rules

Sharing rules are used to extend access to records based on defined criteria or ownership. The project documentation describes Crop records being shared with Farm Manager users with Read/Write access and Buyer records being shared with Sales Department users with Read/Write access.

9. RECORD CREATION AND USER ADOPTION

A Farmer record is created to demonstrate the use of the configured custom objects and relationships. The Farms tab also shows a recently viewed farm record in the configured application.

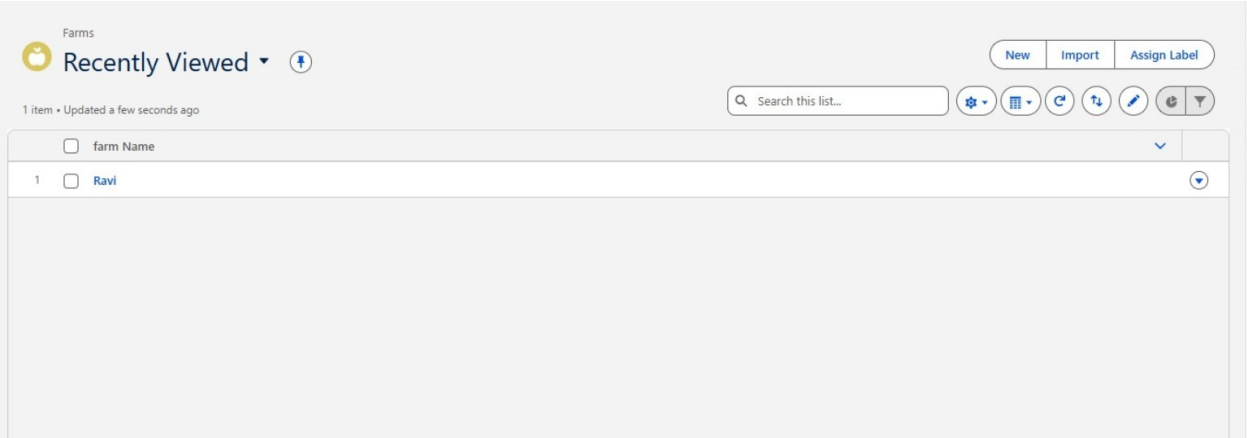


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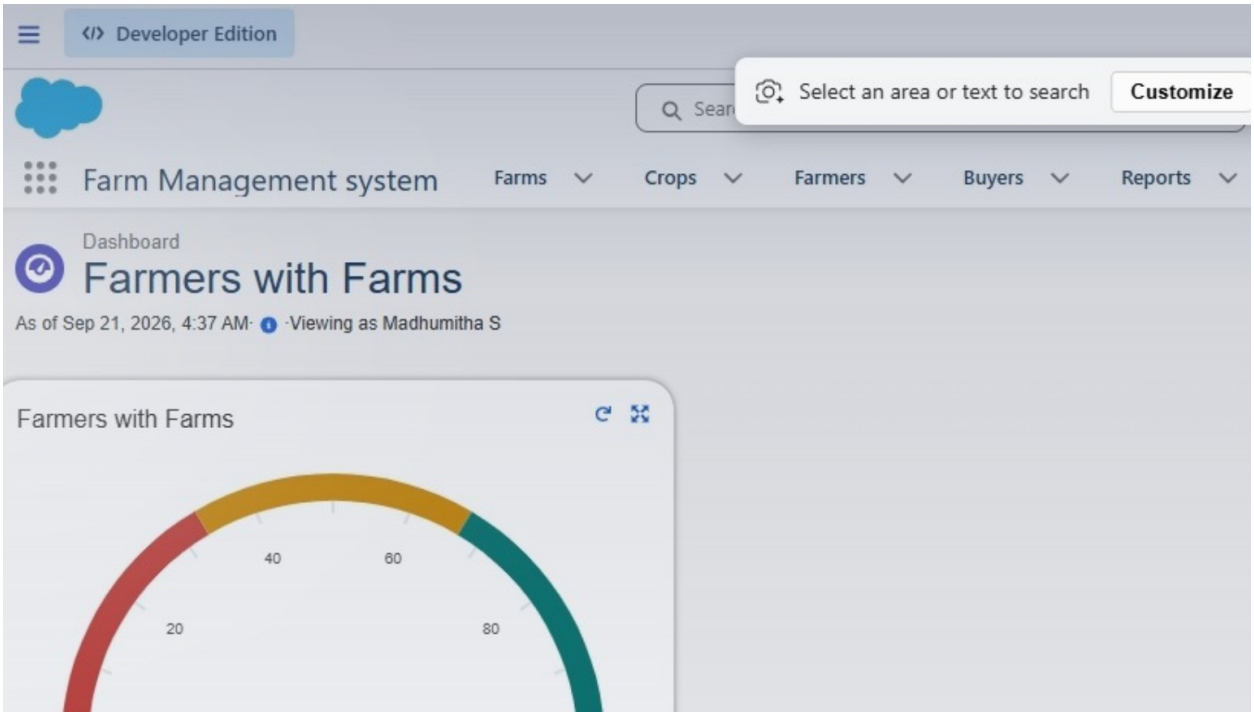


Figure 6. Farm Management System dashboard showing the Farmers with Farms gauge.

10. REPORTS AND ANALYTICS

10.1 Farmers with Farms Report

The project includes a report named “Farmers with Farms”. The displayed report contains Farmer ID, Farmer Name, Farm Name, Location, and Size. The sample record shown in the uploaded evidence is for farmer Ravi and farm Ravi, with location Chennai and size 10.

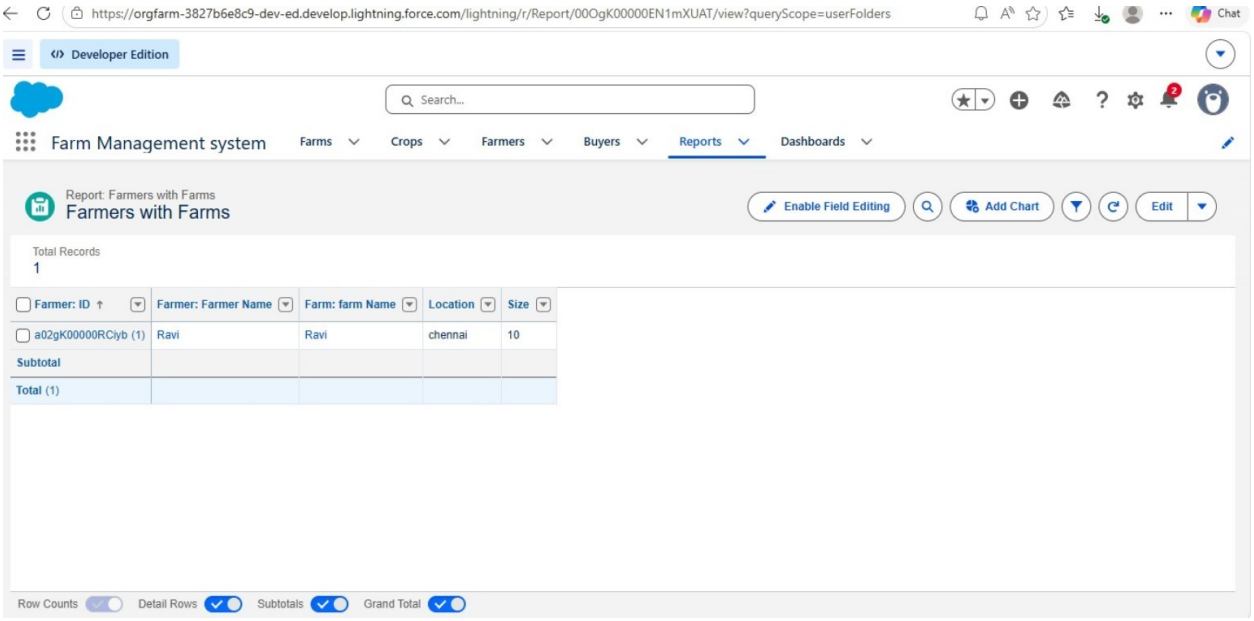


Figure 3. Farmers with Farms report in the Farm Management System.

10.2 Dashboard

A dashboard named “Farmers with Farms” is configured with a gauge component based on the report. The dashboard provides a visual representation of the report data.

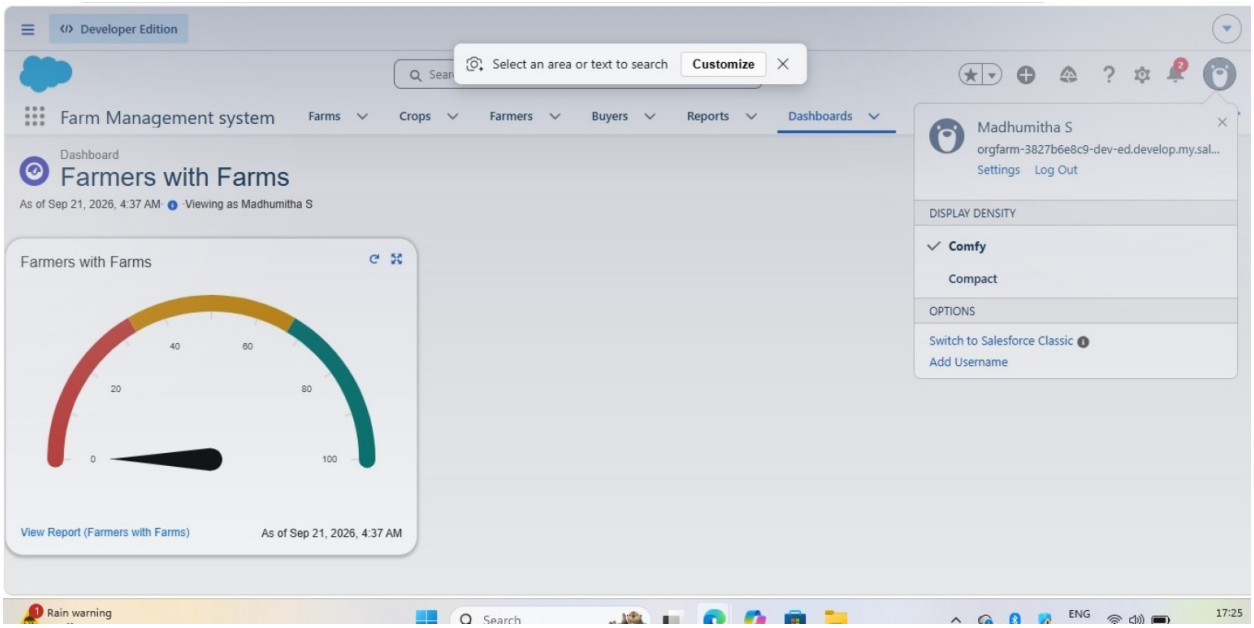


Figure 8. Farm Management System dashboard view with Farmers with Farms.

11. TESTING AND VALIDATION

11.1 Testing Areas

- Creation of Farm, Crop, Farmer, and Buyer records.
- Validation of entered field values.
- Testing of relationships and field dependencies.
- Testing of profile, role, and user access.
- Testing of sharing rules.
- Verification of report generation.
- Verification of dashboard visualization.
- Basic performance and usability review.

11.2 Evidence Summary

The uploaded Salesforce screenshots provide implementation evidence for profiles, custom objects, fields and relationships, custom tabs, records, reports, and dashboards.

12. ADVANTAGES AND LIMITATIONS

12.1 Advantages

- Centralized agricultural data management.
- Reduced paperwork and manual record handling.
- Improved data accuracy and consistency.
- Better monitoring of farms and related records.
- Real-time analytics through reports and dashboards.

12.2 Limitations

- The system depends on internet connectivity.
- Salesforce licensing and platform availability can affect deployment.
- Offline access is limited.

13. CONCLUSION AND FUTURE ENHANCEMENTS

The Farm Management System successfully demonstrates how Salesforce can be used to digitize agricultural data management. The solution centralizes farm, crop, farmer, and buyer information, supports structured data management, provides role-based access, and enables reporting and dashboard analytics. These capabilities can improve operational efficiency, reduce errors, and support data-driven decisions.

13.1 Future Scope

- Dedicated mobile application for farmers and farm managers.
- AI-based crop prediction and recommendation.
- Weather information integration.
- IoT sensor integration for field monitoring.
- Integration with government agricultural schemes.
- Market price forecasting and analytics.

14. APPENDIX – IMPLEMENTATION EVIDENCE

Evidence 1

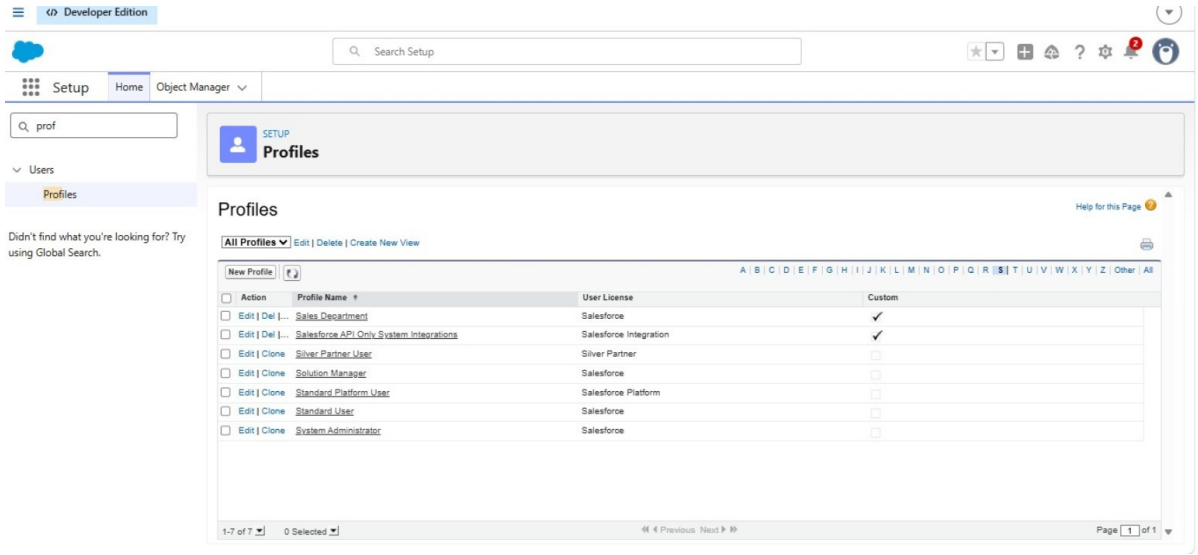


Figure 1. Salesforce Profiles configuration showing the Sales Department profile.

Evidence 2

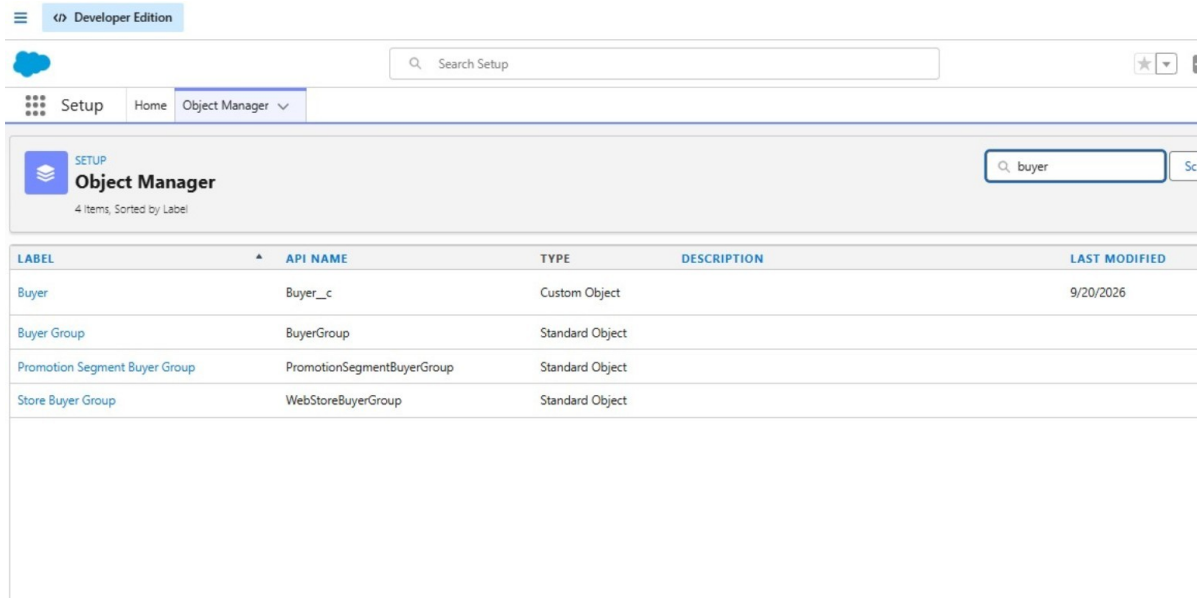


Figure 2. Salesforce Object Manager showing the Buyer custom object.

Evidence 3

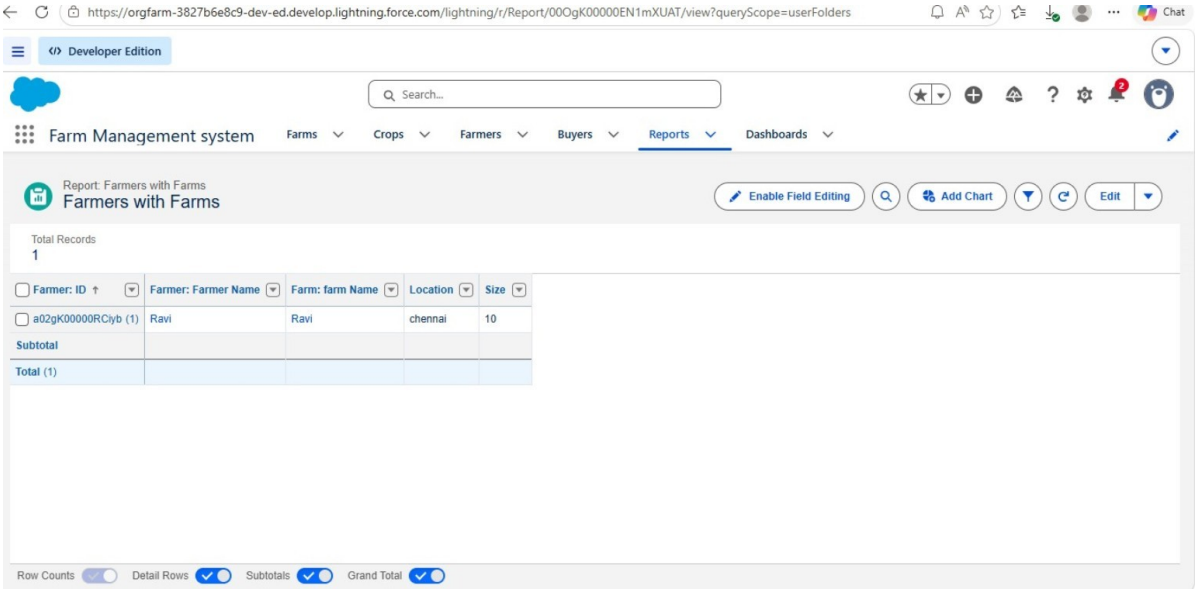


Figure 3. Farmers with Farms report in the Farm Management System.

Evidence 4

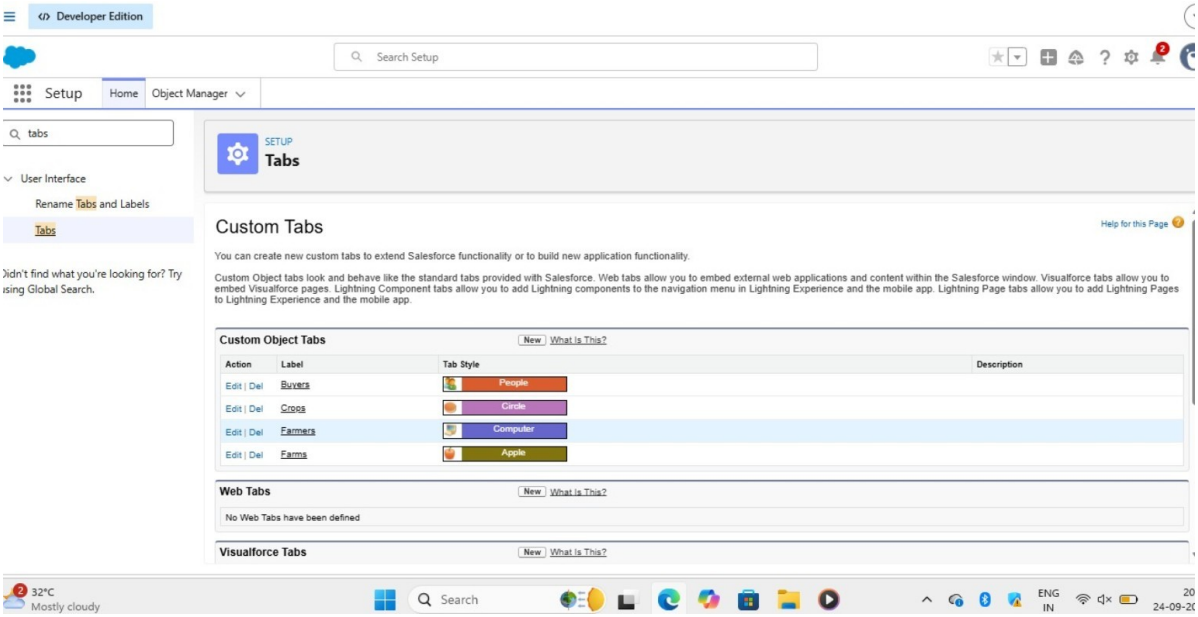


Figure 4. Custom Tabs configuration for Buyers, Crops, Farmers, and Farms.

Evidence 5

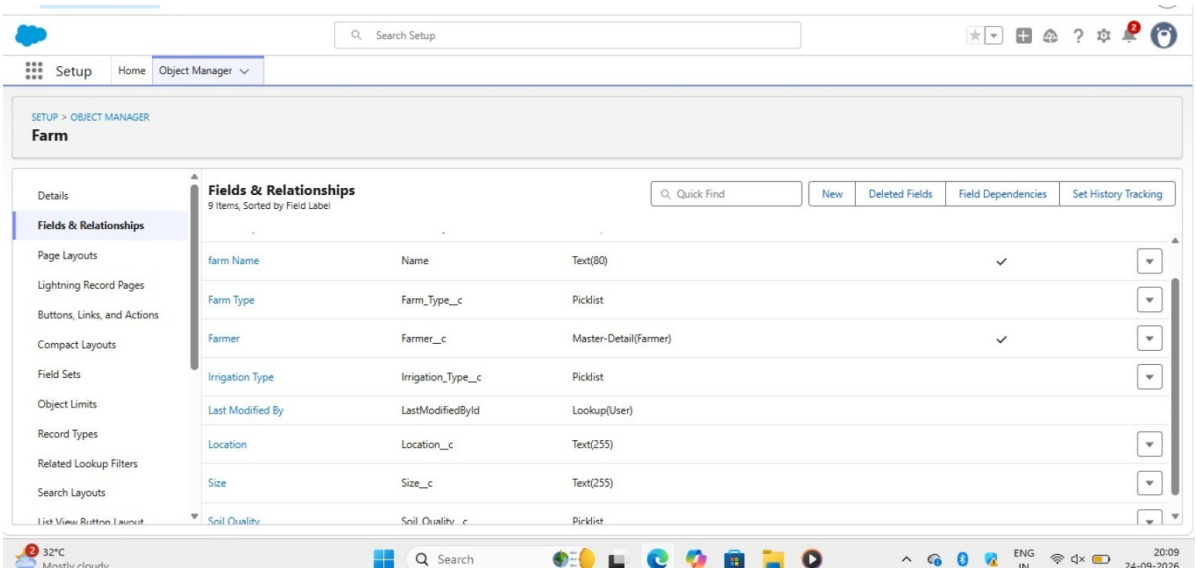


Figure 5. Farm object Fields & Relationships configuration.

Evidence 6

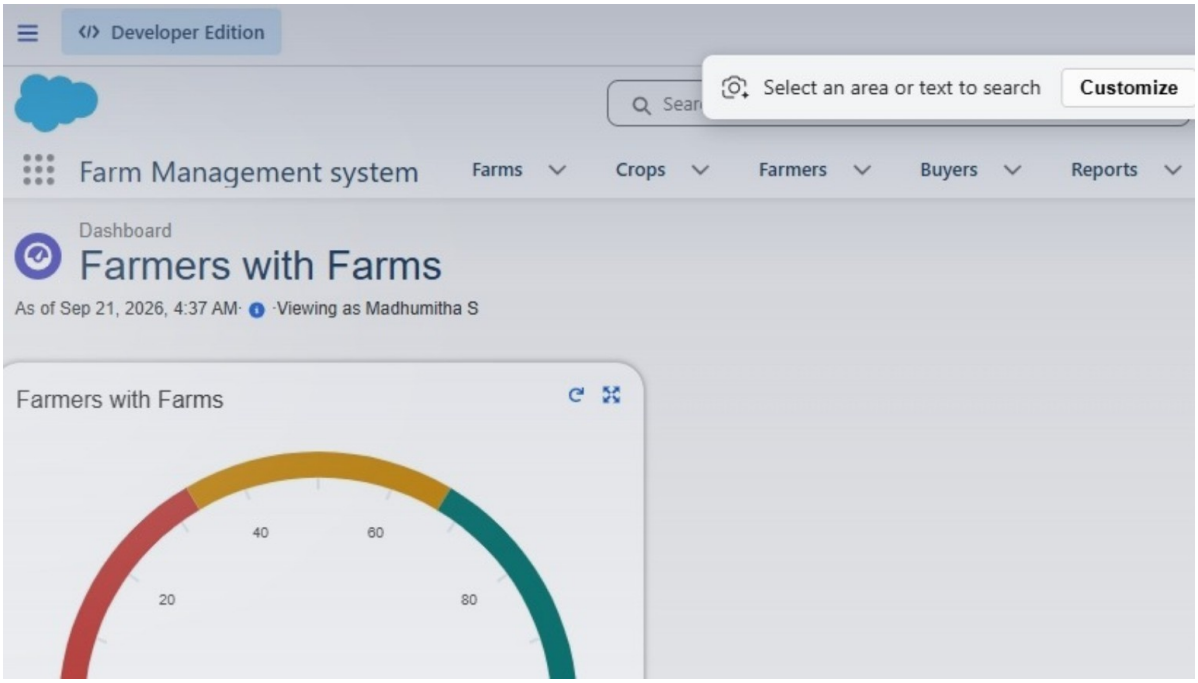


Figure 6. Farm Management System dashboard showing the Farmers with Farms gauge.

Evidence 7

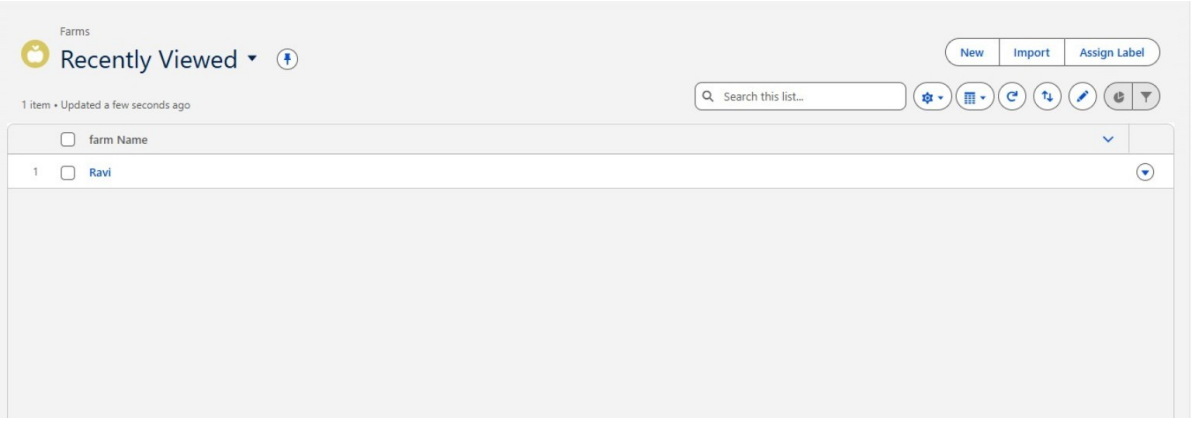


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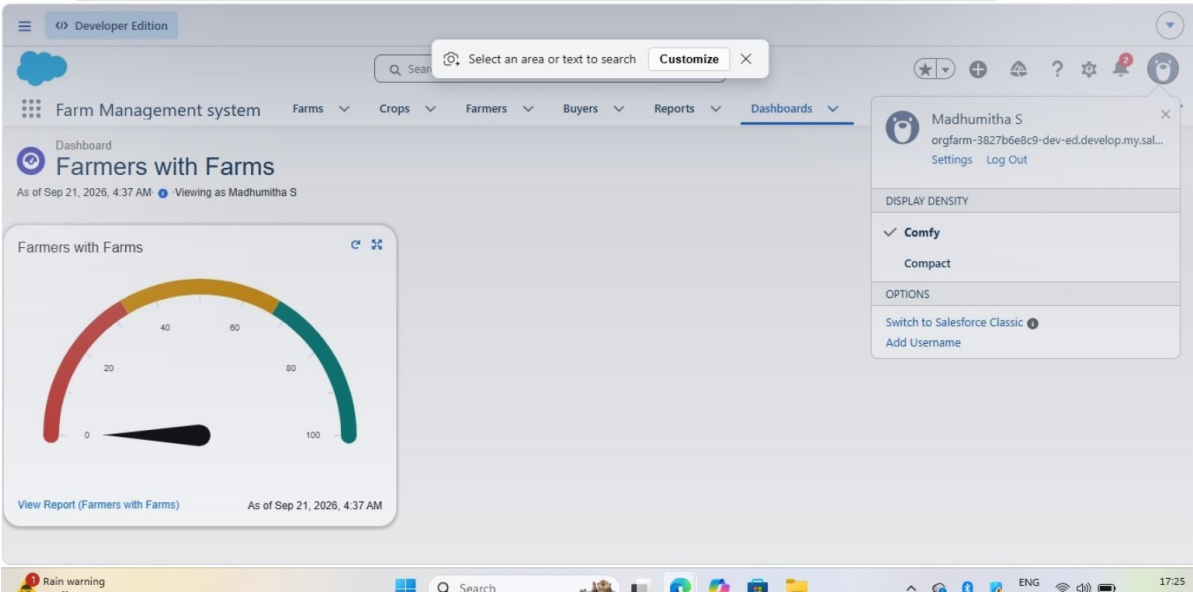


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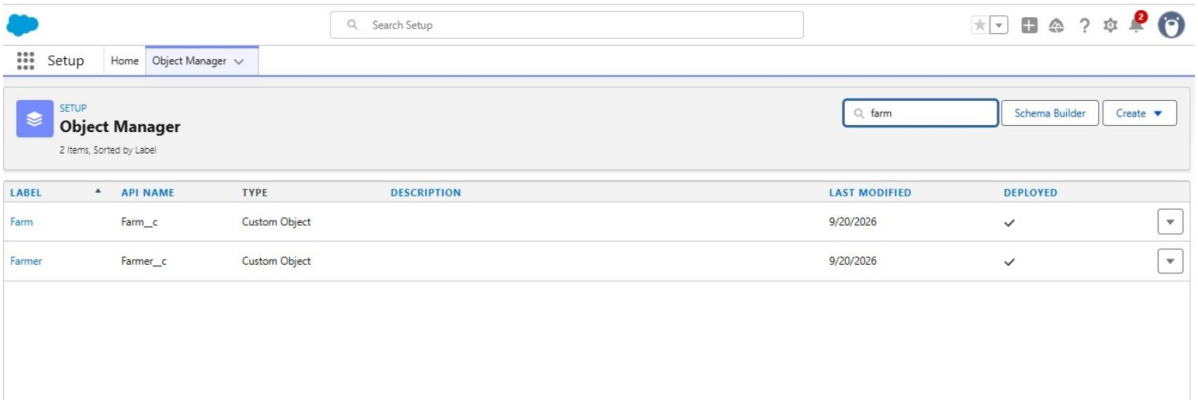


Figure 9. Salesforce Object Manager showing Farm and Farmer custom objects.

Evidence 10

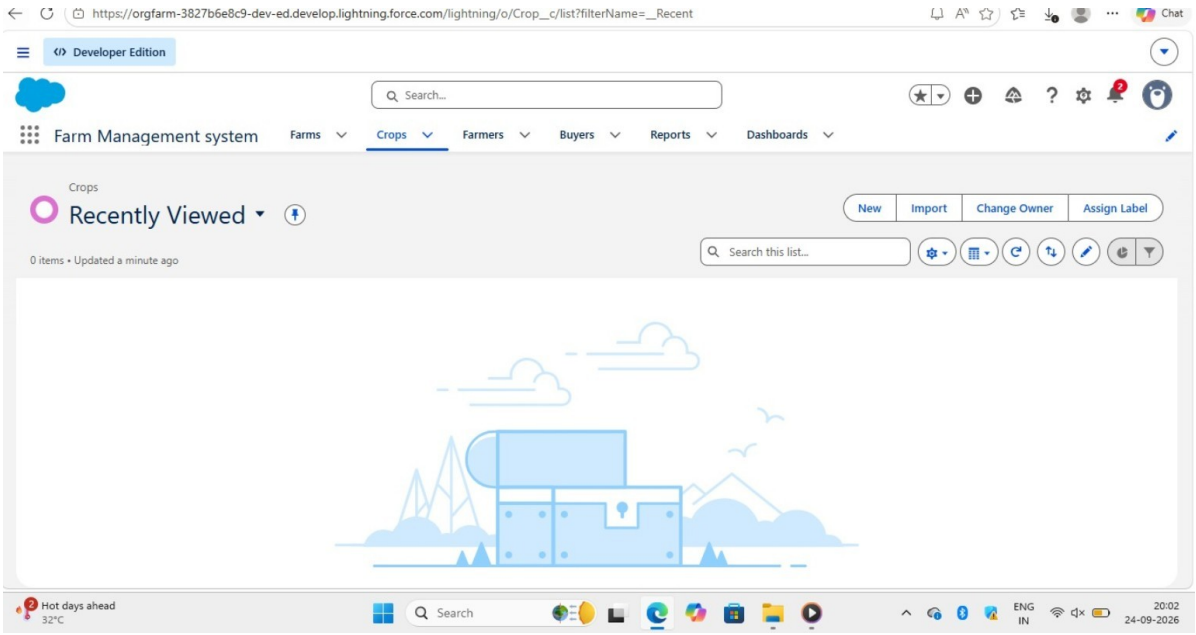


Figure 10. Farm Management System Crops tab.